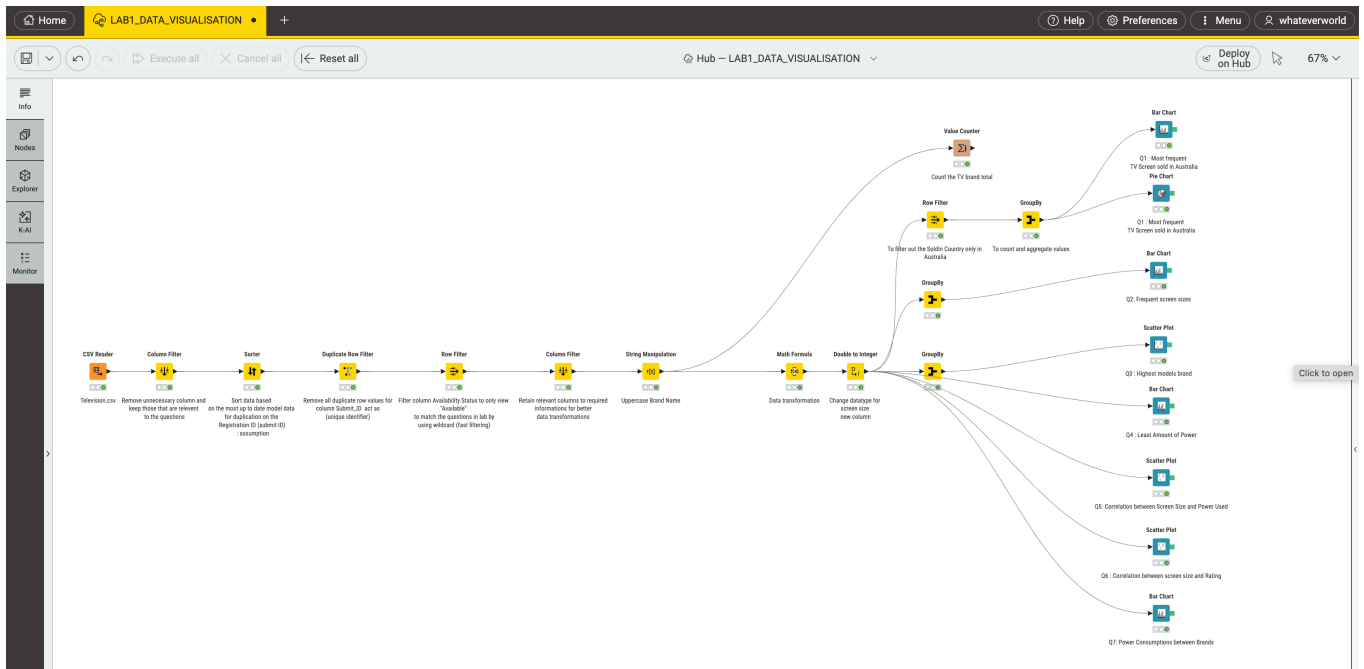


T01 (a): Build Appliance Energy Consumption website >> host on Vercel

Website: <https://data-visualisation-6cbf-git-lab1-whatevershuts-projects.vercel.app>

T01 (b): Data Processing with KNIME



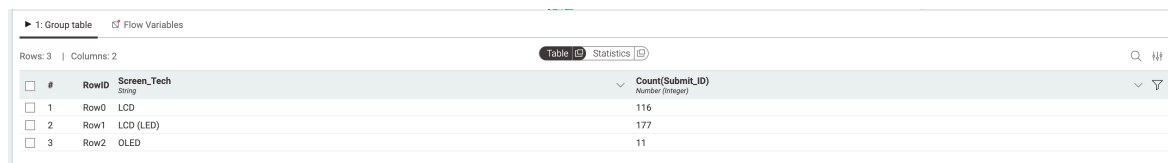
T02: Data Exploration and Basic Chart Types

1. What type of TV screen technologies are currently available in Australia and which are the most frequent?

a. What data is needed to answer this question?

Column Name	Data Type	Description
Screen_Tech	String	Type of screen technology such as LCD, Plasma, OLED
SoldIn	String	This column represents where the product is registered for sale that includes Australia, New Zealand and Fiji
Submit_ID	String	This column represents the unique registration ID record for the product

Answers : LCD(LED) (most frequent)



The screenshot shows a data table with 3 rows and 2 columns. The columns are 'Screen_Tech' (String) and 'Count(Submit_ID)' (Number (Integer)). The rows are labeled 'Row0', 'Row1', and 'Row2'. The data is as follows:

#	RowID	Screen_Tech	Count(Submit_ID)
1	Row0	LCD	116
2	Row1	LCD (LED)	177
3	Row2	OLED	11

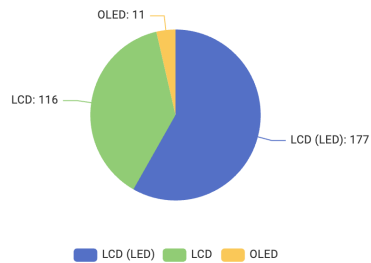
b. What chart type would be good to represent this data and give some reasons?

In general, since the datasets are quite small in numbers both the charts displayed information clearly, but the most suitable chart to show the frequency value easily and fast is pie chart. The reasons are :

1. It shows the total frequency of TV technologies amount clearly by different size of slices
2. The data is distributed well in the circle since the data is small

c. Draw a sketch of the chart

Pie Chart Frequent TV technologies sold in Australia



d. Draw a sketch of the data table required to build the chart

► 1: Group table Flow Variables

Rows: 3 | Columns: 2 Table Statistics Search Filter

#	RowID	Screen_Tech	Count(Submit_ID)
1	Row0	LCD	116
2	Row1	LCD (LED)	177
3	Row2	OLED	11

Row Filter ×

Filter

Criterion 1 ↑ ↓ ✕

Filter column: Soldin Operator: Matches regex

Case matching: Case sensitive Case insensitive

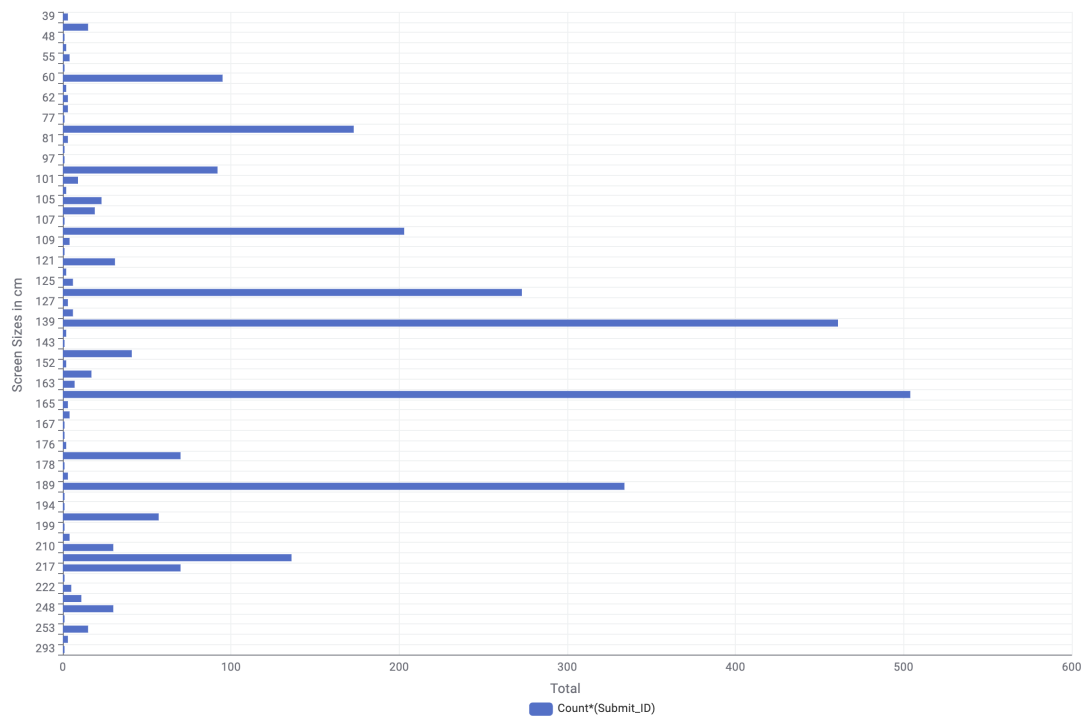
Value: australia

⊕ Add filter criterion

Discard Apply and Execute Apply

2. What screen sizes are currently available, and which are the most frequent?

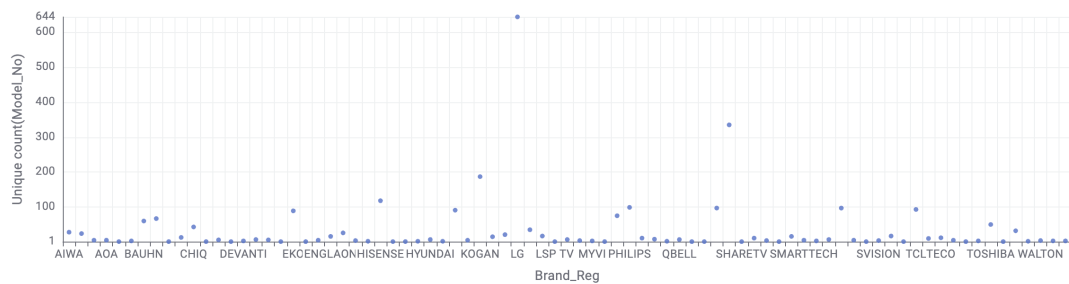
Frequent Screen Sizes



Answer: 164 cm

3. Which brands have the greatest number of different models?

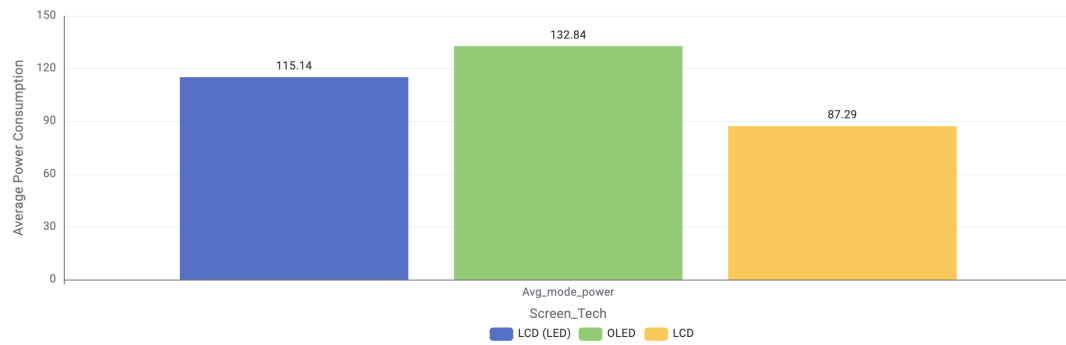
Scatter Plot



Answer: LG

4. Which type of screen technology consumes the least amount of power?

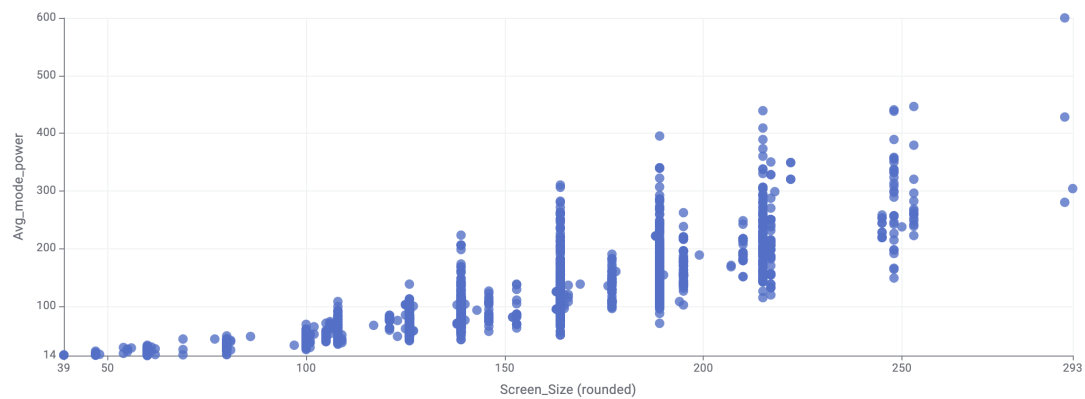
Screen Technology that Uses Least Amount Of Power



Answer: LCD(LED)

5. What is the relationship between screen size and power use?

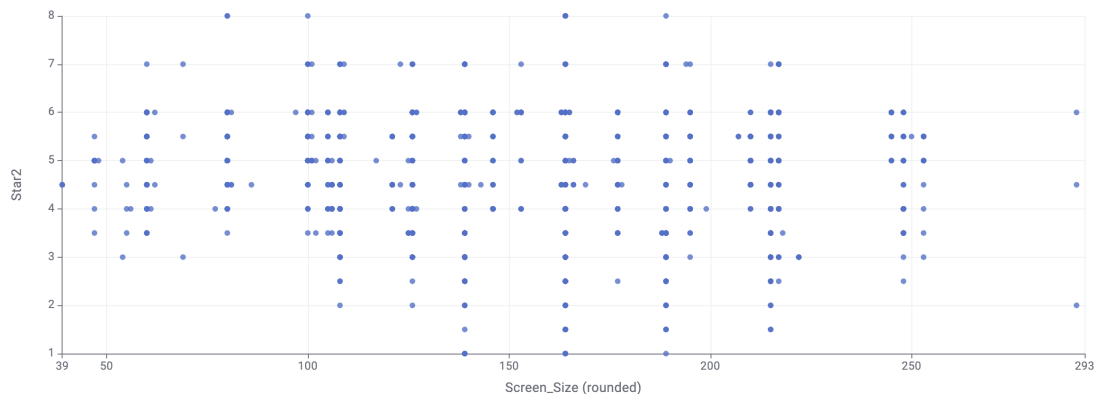
Scatter Plot Correlation between Screen Sizes & Power



Answer: The bigger the screen size the higher the amount of energy used when active

6. What is the relationship between star rating and screen size?

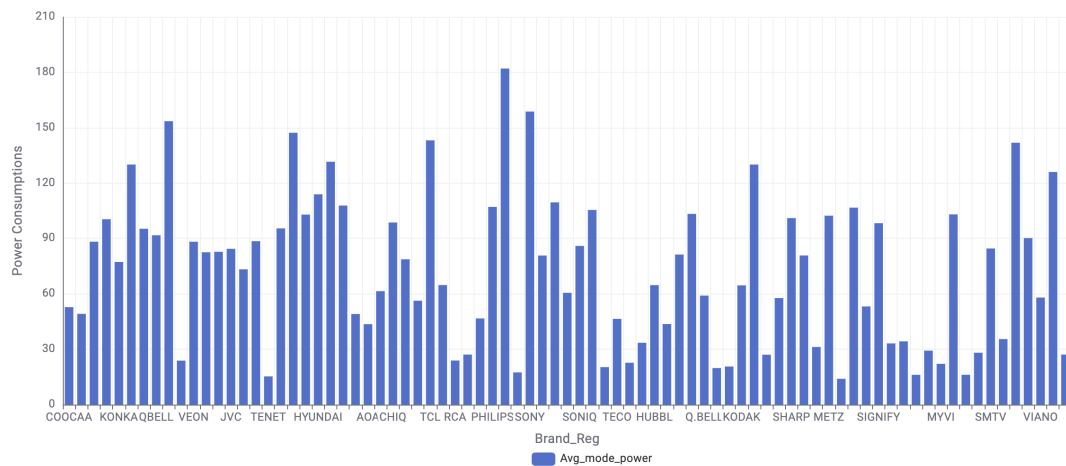
Scatter Plot correlation between Screen Size and Rating



Answer: Small correlation but the hypothesis is : The bigger the screen size the higher the rating

7. Are there differences in power consumption between brands?

Power consumption between Brands



Answer : Yes, evidently the brand EMETE uses the highest amount of power when in use while ARK uses the least power